

Estimation and Approximation

Worksheet 1 (Non-Calculator)

1. W17/qp11/q4

By writing each number correct to 1 significant figure, calculate an estimate for the value of

$$\frac{987.65}{0.0193} \quad [2]$$

2. W17/qp12/q10

By making suitable approximations, calculate an estimate for

$$\frac{40.32 \times \sqrt{35.7}}{2980}$$

Show clearly the approximations you use and give your answer correct to 1 significant figure. [2]

3. S19/qp12/q4

By writing each number correct to one significant figure, estimate the value of

$$\frac{71.8 - 32.4}{0.198^2}$$

[2]

4. W19/qp11/q8

By writing each number correct to 1 significant figure, estimate the value of

$$\frac{59.843^2}{20.13 \times 0.9024}$$

[2]

5. W19/qp12/q8

By writing each number correct to 1 significant figure, estimate the value of

$$\frac{39.864 \times \sqrt{8.987}}{0.6013}.$$

[2]

6. S20/qp12/q5

By writing each number correct to 1 significant figure, estimate the value of

$$\frac{2.78^3}{61.4 \times 0.893}.$$

[2]

7. W20/qp11/q8

By writing each number correct to 1 significant figure, estimate the value of

$$\frac{6013 \times 0.0405}{\sqrt{8.986}}.$$

[2]

8. W20/qp12/q7

By writing each number correct to 1 significant figure, estimate the value of

$$\frac{6.044^2}{212 \times 0.304}.$$

[2]

9. S21/qp11/q10

By writing each number correct to 1 significant figure, estimate the value of

$$\frac{362.4 - 187.2}{52.3}.$$

[2]

10. S21/qp12/q4(b)

By writing each number correct to 1 significant figure, estimate the value of

$$\frac{67.8 + 49.5}{0.187^2}.$$

[2]

Answer Key- Worksheet 1

1. (Initial statement containing 1000 and 0.02) 50 000
2. 0.08
3. 1000
4. 200
5. 200
6. $\frac{1}{2}$
7. 80
8. $\frac{3}{5}$
9. 4
10. 3000



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